

Experiment Design Handbook

Multi-View Entity Resolution:
Validating the ERBOT Method Selection Guide

Xinmin Chu

David Degras-Valabregue

Department of Mathematics, University of Massachusetts Boston

Version 1.0 May 29, 2026

Contents

1	Purpose and Scope	3
2	The Method Selection Guide	3
3	Theoretical Foundations	4
3.1	The ER Pipeline	4
3.2	Three Representation Types	4
3.3	Three Fusion Strategies	4
3.4	Normalization	5
3.5	Weight Learning	5
3.6	NA-Aware Weighted Average	5
3.7	Evaluation Metrics	6
4	Benchmark Datasets	7
5	Research Questions	8
6	Experiment Protocols	8
6.1	Exp 1: Main Accuracy Comparison	8
6.2	Exp 2: Field Ablation on CORA	10
6.3	Exp 3: Scalability	11
6.4	Exp 4: Missingness Robustness	12
6.5	Exp 5: Representation Type	13
7	Validation: Linking Scenarios to Experimental Evidence	14
8	Computational Budget and Resource Planning	14
8.1	Hardware Assumptions	14
8.2	Estimated Run Times	14
8.3	Memory Considerations	15
9	ERBOT Implementation Checklist	15
10	Statistical Analysis Plan	16
10.1	Primary Analysis (Exp 1)	16
10.2	Scalability Analysis (Exp 3)	16
10.3	Missingness Analysis (Exp 4)	16

10.4 Representation Type Analysis (Exp 5)	17
11 Risk Register	17
12 Run Schedule	17
13 Summary	17

1 Purpose and Scope

This handbook is the operational companion to the slide deck *Multi-View Entity Resolution: Three Representation Types and Three Fusion Strategies*. Slide 15 of that deck presents a *Method Selection Guide* — a prescriptive table mapping seven distinct ER situations to a recommended fusion strategy (A, B, or C). The table encodes the central empirical claim of the ERBOT paper:

The winning fusion strategy is predictable from field-level diagnostics (cardinality, missingness, per-field ARI) before any full experiments are run.

Scope. This handbook provides:

1. A precise theoretical rationale for each of the seven selection rules in the guide.
2. A complete, reproducible protocol for five experiments (Exp 1–5) that validate the guide across four benchmark datasets.
3. A mapping between each selection rule, the experiments that test it, and the hypothesized outcome.
4. Computational budget estimates, hardware assumptions, and a run schedule.
5. An implementation checklist for the `erbot` R package functions required before experiments can run.

What the handbook does not cover. Theoretical proofs, R package API documentation, and submission logistics are addressed in `erbot.tex` and `multiview_er_slides.tex`.

2 The Method Selection Guide

Table 1 reproduces and annotates the guide from Slide 15. Each row is treated as a falsifiable hypothesis in Section 7.

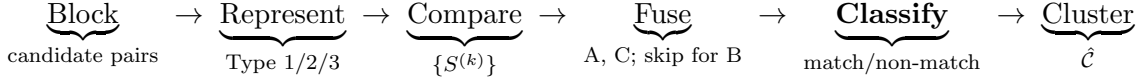
Table 1: Method Selection Guide (Slide 15). Scenarios 1–7 are treated as experimental hypotheses. Exp columns indicate which experiment provides primary evidence.

#	ER Situation	Recommended / Rationale	Primary Exp
S1	Fields are noisy variants of entity identity	Strategy A : fusion averages noise across views	Exp 1, Exp 2
S2	Fields capture orthogonal dimensions	Strategy B : independence of views is a feature	Exp 1, Exp 2
S3	Heterogeneous views, local structure present	Strategy C (SNF): iterative diffusion integrates complementary info	Exp 1
S4	Fields represent different relation types	Strategy C (Multiplex): multilayer graph is the right framing	Exp 1
S5	High missingness (>30%)	Strategy A (NA-aware) or Strategy B : NA formula or field skip handle absence	Exp 4
S6	$n > 50K$, scalability critical	Strategy B : parallel per-field classification	Exp 3
S7	Ground truth available	Strategy A (ARI weights): supervised MKL optimizes weights	Exp 1

3 Theoretical Foundations

3.1 The ER Pipeline

Entity resolution identifies records r_i, r_j that refer to the same real-world entity [Christen, 2012]. The canonical pipeline is:



Key principle. The explicit classification step precedes clustering. Clustering is performed on the binary match graph M , not on raw similarity scores S_{ij} . The decision threshold t is always set by a statistical model (Fellegi–Sunter EM [Fellegi and Sunter, 1969] or supervised ARI), making it a *principled boundary*: auditable and grounded in ER semantics.

3.2 Three Representation Types

Definition 1 (Representation type). *A representation type determines how field values are converted to a numerical form suitable for similarity computation. The three types are parallel alternatives, not a hierarchy.*

Type 1 — Embedding.

Field value $\rightarrow$ dense $\mathbb{R}^d$ vector via a pre-trained LLM or BERT encoder. Similarity: cosine distance in embedding space. Strengths: semantic richness, typo tolerance, no feature engineering. Weakness: black-box, GPU-intensive.

Type 2 — Bag of Words (BoW).

Field value $\rightarrow$ sparse token vector weighted by IDF (unigrams or character n -grams). Similarity: Jaccard or TF-IDF cosine. Strengths: interpretable, fast, no training required. Weakness: order-blind, no semantics.

Type 3 — Field Similarity.

Pair $(i, j) \rightarrow$ scalar score $S_{ij}^{(k)}$ via a domain-specific function: Jaro–Winkler [Winkler, 1990] for short text, exact match for categoricals, absolute difference for dates/numerics. Strengths: domain-precise, NA-aware, sparse-matrix-friendly. Weakness: manual design per field type.

This paper focuses on Type 3. Type 1 and Type 2 serve as baselines in Exp 5 (**RQ5**).

3.3 Three Fusion Strategies

Let $S^{(1)}, \dots, S^{(p)}$ be field-level similarity matrices over the candidate pair set $\mathcal{P} \subset [n]^2$.

Strategy A — Late clustering (MKL family).

$$\{S^{(k)}\} \xrightarrow{w_k} S = \sum_{k=1}^p w_k S^{(k)} \xrightarrow{t} M_{ij} \in \{0, 1\} \xrightarrow{\text{CC}} \hat{\mathcal{C}}_A$$

Fuse first, then classify once. Equivalent to Multiple Kernel Learning (MKL) when $S^{(k)}$ are positive semi-definite kernels [Lanckriet et al., 2004].

Strategy B — Early clustering (consensus).

$$S^{(k)} \xrightarrow{t^{(k)}} M_{ij}^{(k)} \in \{0, 1\} \xrightarrow{\text{CC}} \hat{C}^{(k)} \xrightarrow{\alpha} \hat{C}_B$$

Classify independently per field; combine binary decisions by adaptive majority vote:

$$r_i \sim r_j \iff \frac{1}{p_{ij}} \sum_{k: S_{ij}^{(k)} \neq \text{NA}} M_{ij}^{(k)} \geq \alpha$$

where p_{ij} is the count of non-missing fields for pair (i, j) .

Strategy C — Iterative / hybrid. Two variants are compared:

- **SNF** [Wang et al., 2014]: row-stochastize each $S^{(k)}$ to $P^{(k)}$, iterate until convergence: $P^{(k)} \leftarrow P^{(k)} \cdot \frac{1}{m-1} \sum_{l \neq k} P^{(l)} \cdot (P^{(k)})^\top$. Cluster the fused graph spectrally.
- **Multiplex community detection** [Mucha et al., 2010]: treat each $S^{(k)}$ as a layer $A^{(k)}$ of a multiplex graph. Maximize joint modularity: $Q = \sum_{k, ij} [A_{ij}^{(k)} - \gamma_k P_{ij}^{(k)}] \delta(c_i, c_j)$ via Leiden [Traag et al., 2019]/Louvain [Blondel et al., 2008] with inter-layer coupling ω .

3.4 Normalization

Different fields produce similarities on incompatible scales. Strategy A is directly distorted by scale mismatch; B and C are more robust but not immune. Four normalization methods are compared in Exp 1 (factor: normalization):

Method	Formula	Use case in ER
None	$\tilde{S}^{(k)} = S^{(k)}$	Baseline / already-comparable fields
Min-max	$\tilde{S}_{ij}^{(k)} = \frac{S_{ij}^{(k)} - \min}{\max - \min}$	Bounded $[0, 1]$ similarities (Jaro-Winkler)
z -score	$\tilde{S}_{ij}^{(k)} = \frac{S_{ij}^{(k)} - \mu_k}{\sigma_k}$	Wildly different distributions; date/numeric
Row-stochastic	$\tilde{S}_{ij}^{(k)} = S_{ij}^{(k)} / \sum_j S_{ij}^{(k)}$	Sparse matrices; required by SNF

3.5 Weight Learning

Strategy A requires field weights w_k . Four methods are compared (factor: weight in Exp 1):

Method	When to use	MKL analogy
Equal ($w_k = 1/p$)	No prior knowledge	MKL uniform
Fellegi–Sunter EM	Unsupervised	Unsupervised MKL
ARI-based	Ground truth available	Supervised MKL
Bimodality / variance	Field discriminability	Heuristic MKL

3.6 NA-Aware Weighted Average

For Type 3 with missing fields, Strategy A computes the fused score as:

$$S(i, j) = \frac{\sum_{k: S_{ij}^{(k)} \neq \text{NA}} w_k S_{ij}^{(k)}}{\sum_{k: S_{ij}^{(k)} \neq \text{NA}} w_k}$$

The denominator is adaptive: it sums only over observed fields for pair (i, j) . No imputation is performed. This is the principal contribution of ERBOT to missingness handling, directly supporting scenario S5.

3.7 Evaluation Metrics

Four metrics are used, chosen to provide complementary views (chance-corrected vs. not; pair-level vs. record-level vs. information-theoretic) without redundancy. Table 2 lists the properties of each.

Table 2: Metric properties. “Chance-corrected”: expected value is 0 under random clustering. “Size-bias”: large clusters disproportionately influence the score.

Metric	Space	Chance-corr.	Size-bias	Decomposable?
ARI	Pair	Yes	Moderate	No (scalar)
B-cubed F_1	Record	No	Low	Yes (P, R)
V-measure	Information	No	Low	Yes (H, C)
AMI	Information	Yes	Low	No (scalar)
<i>Pairwise F_1</i>	<i>Pair</i>	<i>No</i>	<i>High</i>	<i>Yes (P, R)</i>

Note on pairwise F_1 . Pairwise F_1 is pair-level and shares the same $\binom{n}{2}$ pair space as ARI. The two are algebraically related and highly correlated in practice ($r > 0.90$ across standard ER benchmarks). Moreover, pairwise F_1 is strongly biased toward large clusters: a cluster of size m contributes $\binom{m}{2}$ pairs, so CORA’s 182 entities dominate while NCVB’s near-singleton entities contribute almost nothing. For these reasons pairwise F_1 is **replaced by V-measure** as the third primary metric. Pairwise F_1 is retained in supplementary tables for comparability with prior ER literature.

ARI (Adjusted Rand Index).

$\text{ARI} \in [-1, 1]$; 1 = perfect, 0 = random [Hubert and Arabie, 1985]. Primary accuracy metric. Also used as the supervised loss for ARI-based weight learning in Strategy A; see Section 6.1 for the nested-CV protocol that prevents target leakage.

B-cubed F_1 . For each record r , let A = predicted cluster of r , B = true cluster of r . Define:

$$P_r = \frac{|A \cap B|}{|A|}, \quad R_r = \frac{|A \cap B|}{|B|}, \quad F_{1,r} = \frac{2P_r R_r}{P_r + R_r}$$

Aggregate by averaging over all records. Handles singletons well; lower size-bias than pairwise F_1 [Bagga and Baldwin, 1998].

V-measure (H, C, V).

Information-theoretic decomposition [Rosenberg and Hirschberg, 2007]:

$$\begin{aligned} H &= 1 - \frac{H(\mathcal{C}|\hat{\mathcal{C}})}{H(\mathcal{C})} \quad (\text{homogeneity: each cluster is pure}) \\ C &= 1 - \frac{H(\hat{\mathcal{C}}|\mathcal{C})}{H(\hat{\mathcal{C}})} \quad (\text{completeness: each entity is contained}) \\ V &= \frac{(1 + \beta^2) H \cdot C}{\beta^2 H + C}, \quad \beta = 1 \end{aligned}$$

Reports H and C separately, revealing whether errors are predominantly splits (low C) or merges (low H). Computed via `ClusterR::external_validation()` or `aricode::NMI()`.

AMI (Adjusted Mutual Information).

Chance-corrected version of NMI [Vinh et al., 2010]:

$$\text{AMI} = \frac{I(\mathcal{C}; \hat{\mathcal{C}}) - \mathbb{E}[I]}{\frac{1}{2}[H(\mathcal{C}) + H(\hat{\mathcal{C}})] - \mathbb{E}[I]}$$

AMI $\in [-1, 1]$; 0 under random clustering. Provides an independent chance-corrected view from information theory; detects when ARI and information content disagree (a sign of severe cluster-size imbalance). Computed via `aricode::AMI()`.

Circular dependency note (ARI as both metric and loss). ARI-based weight learning in Strategy A maximizes ARI on training pairs. Evaluating those same weights by ARI on the *same* fold would overestimate performance. The CV protocol (Section 6.1) uses a nested loop: inner folds tune weights; outer fold evaluates all four metrics on held-out data. This is mandatory; it must be respected in every call to `er_weights(method="ari")`.

Wall-clock time (seconds) and peak resident memory (MB) are additionally recorded for every run in Exp 3.

4 Benchmark Datasets

Table 3 summarizes the four datasets. Each was chosen to exercise a specific axis of variation. Table 4 states the blocking policy for each dataset; blocking is applied only where the full Cartesian pair set exceeds practical memory limits.

Table 3: Benchmark datasets.

Dataset	Records	Entities	Fields	Mode	Primary role
CORA	1,879	182	15	Dedup	Multi-field ablation (Exp 2); RQ1, RQ2
Restaurant	864	754	5	Linkage	Clean small benchmark; all three types (Exp 5)
DBLP-ACM	4,910	2,690	4	Linkage	Controlled missingness injection (Exp 4); RQ2, RQ4
NCVR	100,000	98,521	4	Linkage	Large-scale stress test (Exp 3); RQ3

Table 4: Per-dataset blocking policy. Blocking is a scalability device; it is omitted when all pairs are already feasible.

Dataset	Records	All pairs	After blocking	Policy	Blocking key
Restaurant	864	$\approx 373\text{K}$	—	None (all pairs)	—
CORA	1,879	$\approx 1.76\text{M}$	—	None (all pairs)	—
DBLP-ACM	4,910	$\approx 12\text{M}$	$\approx 1\text{--}2\text{M}$	Block on field	venue
NCVR	100,000	$\approx 5\text{B}$	$< 2\text{M}$	Mandatory	surname prefix (block.size=6)

CORA. 1,879 bibliographic references, 182 true entities (≈ 10.3 references/entity). 15 heterogeneous fields: long text (title, abstract), structured lists (authors), categoricals (venue, publisher), and numerics (year). The highest field count of any benchmark used here; ideal for field ablation (Exp 2) and for testing S1 vs. S2. **Blocking: none.** $\binom{1879}{2} \approx 1.76\text{M}$ pairs; manageable without blocking.

Restaurant. 864 records from Fodors and Zagats restaurant guides, 754 true entities. 5 fields: name, address, city, phone, cuisine. Small and relatively clean; all three representation types are

applicable (Exp 5) because all fields are short text. **Blocking: none.** $\binom{864}{2} \approx 373\text{K}$ pairs; no pruning needed.

DBLP-ACM. 4,910 publications drawn from DBLP and ACM digital library, 2,690 true entities. 4 fields: title, authors, venue, year. Moderate size; used as the primary vehicle for Exp 4 because missingness can be injected in a controlled way without confounding field heterogeneity. **Blocking: on venue.** $\binom{4910}{2} \approx 12\text{M}$ pairs without blocking; blocking on **venue** reduces candidate pairs to $\approx 1\text{--}2\text{M}$.

NCVR (North Carolina Voter Registration). 100,000 records drawn from 10 parties, 98,521 true entities (high duplicate rate). 4 fields: given name, surname, suburb, postcode. Used exclusively for scalability testing (Exp 3). **Blocking: mandatory** (`block.size=6` on surname prefix). Without blocking, $\binom{100000}{2} \approx 5\text{B}$ pairs exceed any feasible memory limit; after blocking, candidate pairs remain below 2M.

5 Research Questions

Six research questions (RQ1–RQ6) drive the experimental program. Table 5 maps each RQ to the guide scenarios it addresses and to the experiment that provides primary evidence.

Table 5: Research questions, associated guide scenarios, and primary experiments.

RQ	Scenarios	Question	Primary Exp
RQ1	S1, S2, S3, S4	Does A, B, or C recover entities more accurately across datasets and field configurations?	Exp 1
RQ2	S1, S2, S5	How do field discriminability and missingness rate affect each strategy? Which degrades most gracefully?	Exp 2, Exp 4
RQ3	S6	How does each strategy scale with n ? Does parallel per-field classification give B an advantage?	Exp 3
RQ4	S1, S3	Does normalization strategy interact with fusion strategy? Does A degrade more under scale mismatch?	Exp 1
RQ5	S1–S4	How does representation type (1, 2, 3) interact with fusion strategy?	Exp 5
RQ6	S1–S7	Can field-level diagnostics predict the winning strategy before experiments run?	Exp 1–5 (meta)

6 Experiment Protocols

6.1 Exp 1: Main Accuracy Comparison

Purpose

Establish which fusion strategy achieves highest accuracy (ARI, B-cubed F_1 , V-measure, AMI) across the full benchmark suite and identify interactions among strategy, weight method, and normalization. Addresses **RQ1**, **RQ4**, and guide scenarios S1–S4, S7.

Factorial Design

Factor	Levels
Dataset	CORA, Restaurant, DBLP-ACM, NCVB
Strategy	Strategy A (late clustering), Strategy B (early consensus), Strategy C (SNF, Multiplex)
Weight / α	A: Equal, Fellegi–Sunter EM, ARI-based, Bimodality; B: $\alpha \in \{0.3, 0.5, 0.7\}$; C: (γ, ω) (see Sec. 6.1)
Normalization	None, row-stochastic, z -score, min-max
Clustering	Connected components (primary); Louvain [Blondel et al., 2008]; HC-Average (comparison)
CV fold	5-fold stratified by entity
Metric	ARI (primary); B-cubed F_1 ; V-measure (H, C); AMI (supplementary)

Condition Counts

Not all factor combinations are meaningful. Table 6 lists the estimable conditions.

Table 6: Exp 1 condition counts per dataset.

Strategy	Conditions (weight $\times$ norm $\times$ clustering)	$\times$ 5 folds
A	$4 \times 4 \times 3 = 48$	240
B	$3 \times 4 \times 1 = 12$	60
C (SNF)	$1 \times 1 \times 1 = 1$	5
C (Multiplex)	$3 \times 3 \times 1 = 9$	45
Total per dataset	70 conditions	350 runs
Total (4 datasets)		1,400 runs

Notes. SNF requires row-stochastic normalization; no other normalization applies. Multiplex conditions: $\gamma \in \{0.5, 1.0, 1.5\}$, $\omega \in \{0.1, 0.5, 1.0\}$. Strategy B clustering is implicit in the consensus vote; CC is used for single-field sub-problems.

Protocol

Nested CV for ARI-based weights. When `er_weights(method="ari")` is used inside Strategy A, a nested loop is required to avoid target leakage: outer fold o is held out for evaluation; inner 4-fold CV on the remaining data tunes the weights. The outer-fold ARI is then computed on o with the weights fixed. All other weight methods (Equal, FS-EM, Bimodality) use standard (non-nested) 5-fold CV.

1. **Preprocessing.** Load each dataset with `er_load()`. Apply the per-dataset blocking policy (see Section 4 and Table 4): Restaurant and CORA use all pairs (no blocking); DBLP-ACM blocks on `venue` [Papadakis et al., 2020]; NCVB blocks on surname prefix (`block_size=6`). Record candidate pair counts after this step.
2. **5-fold split.** Split at the *entity* level (not the record level) to prevent information leakage across folds. Use `er_split(k=5, entity_disjoint=TRUE)`.
3. **Representation.** Compute Type-3 similarities with `er_similarity()`. Store the list $\{S^{(k)}\}_{k=1}^p$ as a named list object.
4. **Normalization.** Apply each normalization method with `er_normalize(method=)`. Four normalized versions of $\{S^{(k)}\}$ are created per fold.
5. **Strategy A — fuse then classify.** For each (weight method, normalization) combination:

- (a) Learn weights with `er_weights(method=)`. For `method="ari"`, use nested inner CV (see above).
 - (b) Fuse with `er_combine(weights=)`.
 - (c) Set threshold with `er_classify(threshold=)`; threshold estimated by Fellegi–Sunter EM or ARI grid search.
 - (d) Cluster with `er_cluster(method=)`.
6. **Strategy B — classify per field then consensus.** For each $(\alpha, \text{normalization})$ combination:
- (a) For each field k : classify with `er_classify()`, producing $M^{(k)}$.
 - (b) Aggregate: `er_consensus(alpha=)`.
7. **Strategy C — iterative fusion.**
- (a) SNF: `er_snf(K=20, t=20)` (top- K network; t iterations).
 - (b) Multiplex: `er_multiplex(gamma=, omega=)`.
 - (c) Cluster the fused graph with spectral / Leiden.
8. **Evaluate.** Compute ARI, B-cubed F_1 , V-measure (H, C) , and AMI via `er_eval()`. Log wall-clock time and peak memory.
9. **Aggregate.** Compute mean $\pm$ standard deviation across 5 folds. Run a two-way ANOVA (strategy $\times$ normalization) on ARI.

Primary Output

A table of mean ARI $\pm$ SD and V-measure (H, C) per (dataset, strategy) combination, analogous to the result table of the paper. A heat map of strategy $\times$ normalization interaction effects on ARI. Supplementary table: B-cubed F_1 and AMI for completeness.

Hypotheses Tested

Hypothesis 1 (S1). *On CORA (heterogeneous fields that jointly identify citations), [Strategy A](#) achieves the highest ARI when weights are learned by ARI-based or Fellegi–Sunter EM.*

Hypothesis 2 (S7). *On datasets with ground truth, ARI-based weighting improves over equal weighting by ≥ 0.05 ARI points.*

Hypothesis 3 (S4: normalization). *Strategy A ARI degrades by > 0.10 when normalization is omitted on datasets with heterogeneous field scales; B and C degrade less.*

6.2 Exp 2: Field Ablation on CORA

Purpose

Identify which fields drive accuracy for each strategy. Quantify how much ARI drops when each of CORA’s 15 fields is removed one at a time. Addresses **RQ2** and scenarios S1 (complementary fields) vs. S2 (orthogonal fields).

Design

Factor	Levels
Field removed	$\{\text{none}\} \cup \{\text{each of the 15 CORA fields}\} = 16$ conditions
Strategy	A (best configuration from Exp 1), B ($\alpha = 0.5$), C (SNF)
CV fold	5-fold stratified by entity
Total runs	$16 \times 3 \times 5 = 240$

Protocol

1. Use the best (weight, normalization, clustering) configuration from Exp 1 for Strategy A.
2. For each field $f \in \{1, \dots, 15\}$: set $S^{(f)} = \text{NA}$ (all pairs missing for that field).
3. Run the three strategies under the ablated field set.
4. Record ARI drop: $\Delta\text{ARI}(f, \text{strategy}) = \text{ARI}(\text{full}) - \text{ARI}(\text{without } f)$.

Primary Output

A 15×3 table of ΔARI and ΔC (completeness drop) values, one row per field and one column per strategy. $\Delta C = C(\text{full}) - C(\text{without } f)$ reveals whether removing a field causes splits (entities scattered across clusters), which ΔARI alone does not distinguish from merges. A bar chart showing per-field ΔARI and ΔC side-by-side for each strategy.

Diagnostic Use

The ΔARI profile distinguishes S1 from S2:

- S1 (noisy variants): ΔARI is small and roughly equal across fields for Strategy A; removing any single field barely hurts.
- S2 (orthogonal): ΔARI is large and unequal; one or two fields dominate; Strategy B is more robust to field removal.

Hypothesis 4 (S2). *Fields with high individual ARI (e.g., phone number in Restaurant) produce a ΔARI profile concentrated on those fields; Strategy B matches or exceeds Strategy A ARI under such orthogonal field structure.*

6.3 Exp 3: Scalability

Purpose

Measure how wall-clock time and peak memory scale with dataset size n . Test whether Strategy B’s parallelism confers a practical advantage at $n > 50\text{K}$ (guide scenario S6). Addresses **RQ3**.

Design

Factor	Levels
n (NCVR subset)	{1K, 5K, 10K, 25K, 50K, 100K}
Strategy	A (FS-EM weights, row-stoch), B ($\alpha = 0.5$), C (SNF)
Execution	Sequential; parallel B with <code>parallel::mclapply</code>
Replication	3 independent runs per condition (to estimate variance)
Total conditions	$6 \times 3 = 18$; total runs = 54

Protocol

1. Draw NCVR subsets stratified by entity to preserve duplication rate.
2. Apply standard blocking (`block.size=6`) to each subset. Record candidate pair count $|\mathcal{P}|$ for each n .
3. Wrap each strategy run in `system.time()` and `pryr::mem_used()` (before/after).
4. Run Strategy B in two modes: sequential and parallel (`parallel::mclapply` on all $p = 4$ fields).

5. Fit log-linear regression: $\log(\text{time}) = \alpha + \beta \log(n)$ per strategy. Report estimated β (complexity exponent).

Primary Output

A log-log plot of wall-clock time vs. n for A, B (seq.), B (par.), C. A secondary plot of peak memory vs. n .

Computational Cost Estimate for Exp 3

n	$ \mathcal{P} $ (est.)	Time per run (A)	Time per run (B par.)
1K	$\approx 5\text{K}$	$< 5\text{s}$	$< 3\text{s}$
10K	$\approx 100\text{K}$	$\approx 30\text{s}$	$\approx 15\text{s}$
50K	$\approx 1\text{M}$	$\approx 8\text{min}$	$\approx 3\text{min}$
100K	$< 2\text{M}$	$\approx 25\text{min}$	$\approx 8\text{min}$

Hypothesis 5 (S6). *At $n = 100\text{K}$, parallel Strategy B is at least $3\times$ faster than Strategy A, while achieving ARI within 0.03 of Strategy A’s best.*

6.4 Exp 4: Missingness Robustness

Purpose

Quantify how ARI degrades as fields are masked at increasing rates. Test whether the NA-aware weighted average in Strategy A and the field-skip mechanism in Strategy B are more robust than naïve imputation (zero-imputation baseline). Addresses **RQ2** and guide scenario S5.

Design

Factor	Levels
Dataset	DBLP-ACM (primary); CORA (secondary)
Missingness rate ρ	$\{0\%, 10\%, 20\%, 30\%, 40\%, 50\%\}$ per field
Missing mechanism	MCAR (random); MAR (missingness correlated with field value)
Strategy	A (NA-aware, row-stoch, FS-EM); B ($\alpha = 0.5$); Baseline: A with zero-imputation
CV fold	5-fold
Total runs	$6 \times 2 \times 3 \times 5 = 180$ (DBLP-ACM); ≈ 180 (CORA)

Missingness Injection Procedure

1. Start from the complete-data similarity matrices $\{S^{(k)}\}_{k=1}^p$.
2. **MCAR**: independently mask each entry $S_{ij}^{(k)}$ to NA with probability ρ .
3. **MAR**: mask $S_{ij}^{(k)}$ to NA with probability proportional to $1 - S_{ij}^{(k)}$ (low-similarity pairs more likely missing), normalized so that marginal rate equals ρ .
4. Apply masking on the training fold only; test fold uses complete data for evaluation.

NA-Aware vs. Imputation Comparison

- **NA-aware** (ERBOT default, Section 3.6): adaptive denominator; no imputation.
- **Zero-imputation** (baseline): replace NA with 0 before computing weighted average.
- **Mean-imputation** (optional baseline): replace NA with the mean pairwise similarity for field k .

Primary Output

A line chart of ARI vs. ρ for each strategy and imputation method, faceted by missing mechanism (MCAR vs. MAR). Secondary charts: B-cubed F_1 vs. ρ ; V-measure homogeneity H and completeness C to reveal whether missingness causes merges (low H) or splits (low C); AMI vs. ρ as a chance-corrected information-theoretic cross-check, flagging any strategy $\times$ mechanism interaction that ARI misses.

Hypothesis 6 (S5). *At $\rho = 30\%$ MCAR on DBLP-ACM, the NA-aware Strategy A degrades by ≤ 0.05 ARI points relative to $\rho = 0\%$, while zero-imputation degrades by ≥ 0.15 ARI points.*

6.5 Exp 5: Representation Type

Purpose

Compare the three representation types (Type 1 Embedding, Type 2 BoW, Type 3 Field Similarity) under a fixed Strategy A framework. Determines when investing in Type 1 embeddings pays off relative to the cheaper Type 2 and Type 3 alternatives. Addresses **RQ5**.

Design

Factor	Levels
Dataset	Restaurant (all three types applicable); DBLP-ACM
Representation type	Type 1 (BERT <code>bert-base-uncased</code> embeddings, cosine); Type 2 (character 3-gram TF-IDF, cosine); Type 3 (Jaro-Winkler / exact / numeric diff)
Strategy	A fixed (FS-EM weights, min-max normalization, CC)
CV fold	5-fold
Total runs	$3 \times 2 \times 5 = 30$

Type 1 Implementation Details

- Model: `bert-base-uncased` via the `text` R package (calls the Python `transformers` library).
- Pooling: CLS token embedding, dimension $d = 768$.
- Similarity: cosine distance in $\mathbb{R}^{768}$.
- Hardware: NVIDIA GPU recommended; CPU fallback available (slow).
- Caching: embeddings are computed once per dataset split and saved to `.rds` files to avoid recomputation.

Type 2 Implementation Details

- Tokenization: character 3-grams (or word unigrams for long fields).
- Weighting: TF-IDF with IDF computed on the training split only (no test-set leakage).
- Similarity: cosine via `text2vec::sim2()` [Selivanov et al., 2020].

Primary Output

A bar chart comparing ARI and V-measure across the three types per dataset. A table of ARI $\pm$ SD and AMI for each (dataset, type) pair. Secondary: wall-clock time per type to quantify the embedding overhead.

Hypothesis 7 (RQ5). *On Restaurant (short clean text fields), Type 1 (BERT) improves over Type 3 (Jaro-Winkler) by ≥ 0.05 ARI points. On DBLP-ACM (structured bibliographic fields), the improvement is < 0.03 ARI points and is not statistically significant.*

7 Validation: Linking Scenarios to Experimental Evidence

Table 7 provides a row-by-row account of how each scenario in the Method Selection Guide is tested, what outcome would confirm or refute it, and which experiment supplies that evidence.

Table 7: Validation map: guide scenarios, experiments, and expected outcomes.

S#	Situation	Evidence from	Confirming outcome
S1	Noisy variants	Exp 1 (CORA), Exp 2	$A \geq B$ by ≥ 0.05 ; ablation Δ ARI spread uniformly across fields
S2	Orthogonal dimensions	Exp 1 (Restaurant), Exp 2	$B \geq A$ by ≥ 0.03 ; ablation Δ ARI concentrated on 1–2 dominant fields
S3	Heterogeneous local structure	Exp 1 (CORA, SNF)	C-SNF matches or beats A by ≥ 0.02 on CORA
S4	Different relation types	Exp 1 (Multiplex)	C-Multiplex improves over A on multi-relational fields
S5	High missingness	Exp 4 ($\rho = 30\%$, 50%)	NA-aware A and B degrade ≤ 0.05 ; zero-impute A degrades ≥ 0.15
S6	Large n	Exp 3 ($n = 100K$)	B (parallel) $\geq 3\times$ faster than A; ARI within 0.03
S7	Ground truth available	Exp 1 (ARI weights)	ARI-based $\geq$ equal by ≥ 0.05 ARI points

Meta-analysis (RQ6). After Exp 1–5, a secondary logistic regression model is fit:

$$\Pr[\text{strategy A wins}] = \sigma(\beta_0 + \beta_1 \bar{\text{ARI}}_{\text{per-field}} + \beta_2 \rho + \beta_3 \log n + \beta_4 \text{CV}_{\text{field-ARI}})$$

where $\text{CV}_{\text{field-ARI}}$ is the coefficient of variation of per-field ARI scores (high CV indicates orthogonal fields, favoring B). If $\beta_1 > 0$, $\beta_3 < 0$, and $\beta_2 < 0$ (or interaction) all reach significance at $p < 0.05$, the predictability claim of RQ6 is supported.

8 Computational Budget and Resource Planning

8.1 Hardware Assumptions

Component	Specification
CPU	8-core, 3.5 GHz (experiments parallelized by field)
RAM	32 GB (NCVR at $n = 100K$ requires ≈ 16 GB peak)
GPU	NVIDIA GPU with ≥ 8 GB VRAM (Exp 5 Type 1 only)
Storage	50 GB (intermediate similarity matrices, fold caches)

8.2 Estimated Run Times

Parallelization strategy. Exp 1 with NCVR dominates the budget. The 350 NCVR runs are independent across conditions; batching them with `parallel::mclapply` over 8 cores reduces wall time to ≈ 6 hours. Similarly, Exp 1 non-NCVR conditions can be parallelized to < 2 hours. *Realistic total with full parallelism on 8 cores: ≈ 12 –15 hours.*

Table 8: Estimated total wall-clock time per experiment, sequential execution on the specified hardware.

Exp	Total runs	Avg. per run	Total (est.)
Exp 1 (excl. NCVR)	1,050	45s	≈ 13 hr
Exp 1 (NCVR)	350	8 min	≈ 47 hr
Exp 2 (CORA)	240	10s	≈ 40 min
Exp 3 (scalability)	54	varies	≈ 6 hr
Exp 4 (missingness)	360	60s	≈ 6 hr
Exp 5 (types)	30	10 min (T1) / 30s (T2/T3)	≈ 2 hr
Grand total			≈ 74 hr

8.3 Memory Considerations

- Similarity matrix $S^{(k)}$ for $|\mathcal{P}| = 2\text{M}$ pairs (NCVR): $2\text{M} \times 8$ bytes = 16 MB per field. With $p = 4$ fields and 4 normalized copies: ≈ 256 MB.
- SNF intermediate matrices: $O(n_{\text{block}}^2)$ per block; managed by `er_snf()`’s block-sparse implementation.
- BERT embeddings (Exp 5, Restaurant): $864 \times 768 \times 4$ bytes ≈ 2.5 MB per field; negligible.

9 ERBOT Implementation Checklist

Table 9 lists the `erbot` R package functions required for each experiment. Functions marked $\circ$ (not yet implemented) are prerequisites before the corresponding experiment can run.

Table 9: ERBOT function implementation status and experiment dependencies.

Function	Purpose	Status	Used in
<code>er_load()</code>	Dataset ingestion	✓	All
<code>er_block()</code>	Standard blocking	✓	DBLP-ACM, NCVR
<code>er_split()</code>	Entity-disjoint CV split	✓	All
<code>er_similarity()</code>	Type-3 field similarities	✓	All
<code>er_weights()</code>	Weight learning (A/B)	✓	Exp 1, 4
<code>er_combine()</code>	Weighted fusion (A)	✓	Exp 1, 4
<code>er_consensus()</code>	Majority-vote fusion (B)	✓	Exp 1–5
<code>er_cluster()</code>	CC / Louvain / HC	✓	Exp 1–5
<code>er_eval()</code>	ARI, B-cubed, V-measure, AMI	✓	All
<code>er_normalize()</code>	Row-stoch / z / minmax	$\circ$	Exp 1, 4
<code>er_classify()</code>	FS-EM / ARI threshold	$\circ$	Exp 1–5
<code>er_snf()</code>	SNF iterative fusion (C)	$\circ$	Exp 1
<code>er_multiplex()</code>	Multiplex community (C)	$\circ$	Exp 1
<code>er_bow()</code>	Type-2 BoW similarities	$\circ$	Exp 5
<code>er_embed()</code>	Type-1 BERT embeddings	$\circ$	Exp 5
<code>er_compare_paradigms()</code>	Full strategy comparison	$\circ$	All

Implementation priority. For a minimal Exp 1 that covers scenarios S1–S4 and S7, the required unimplemented functions are: `er_normalize()`, `er_classify()`, and `er_snf()`. These three should be implemented first, in that order. `er_multiplex()` can follow; `er_bow()` and `er_embed()` are needed only for Exp 5.

`er_normalize()` Specification

`er_normalize(S_list,`

```

        method = c("row_stochastic", "minmax", "zscore", "none"),
        na.rm = TRUE)
# Returns: named list of normalized matrices, same structure as S_list.
# Handles NA entries: NA rows excluded from row-sum in row_stochastic;
# NA entries excluded from mean/sd in zscore.

```

er_classify() Specification

```

er_classify(S, method = c("fellegi_sunter", "ari", "fixed"),
            threshold = NULL, truth = NULL, ...)
# Returns: sparse binary matrix M (match = 1).
# method = "fellegi_sunter": EM on score distribution mixture;
#   returns threshold t as attribute(M, "threshold").
# method = "ari": grid search over t, maximize ARI against truth.
# method = "fixed": use supplied threshold directly.

```

er_snf() Specification

```

er_snf(S_list, K = 20, t = 20, alpha = 0.5)
# S_list: list of p similarity matrices (same pairs, possibly sparse).
# K: number of nearest neighbours for the KNN graph.
# t: number of diffusion iterations.
# alpha: self-loop weight in the diffusion update.
# Returns: fused similarity matrix, same dimensions as S_list[[1]].

```

10 Statistical Analysis Plan

10.1 Primary Analysis (Exp 1)

For each dataset, fit a linear mixed-effects model:

$$\text{ARI}_{ijk} = \mu + \underbrace{\alpha_i}_{\text{strategy}} + \underbrace{\beta_j}_{\text{norm}} + \underbrace{(\alpha\beta)_{ij}}_{\text{interaction}} + \underbrace{b_k}_{\text{fold (random)}} + \varepsilon_{ijk}$$

where fold $k \sim \mathcal{N}(0, \sigma_b^2)$ is a random effect. Use `lme4::lmer()` in R. Report:

- Fixed-effect F -tests for strategy and normalization main effects.
- Pairwise Tukey post-hoc comparisons across strategies.
- Fold-level variance component $\hat{\sigma}_b^2$ (estimate of dataset split sensitivity).

10.2 Scalability Analysis (Exp 3)

Fit the power-law model $T_n = a \cdot n^\beta$ by log-linear regression

$$\log T_n = \log a + \beta \log n + \varepsilon$$

per strategy. Report $\hat{\beta} \pm \text{SE}$ and 95% confidence intervals. Test $H_0 : \beta_B = \beta_A$ using a Wald test; rejection supports S6.

10.3 Missingness Analysis (Exp 4)

Report ARI as a function of ρ for each (strategy, mechanism) combination. Test for significant ARI degradation at $\rho = 0.3$ and $\rho = 0.5$ using paired t -tests (paired over folds) with Bonferroni correction for multiple comparisons.

10.4 Representation Type Analysis (Exp 5)

One-way ANOVA on ARI over the three representation types, separately per dataset. Report effect sizes (Cohen’s d) for Type 1 vs. Type 3 and Type 2 vs. Type 3.

11 Risk Register

Risk	Likelihood	Mitigation
NCVR at $n = 100K$ exceeds 2M pair limit after blocking	Medium	Reduce block_size to 5 (tested feasible in prior runs); or subsample to 50K for full Exp 1, use 100K only for Exp 3.
SNF does not converge within $t = 20$ iterations	Low	Monitor Frobenius norm $\ P_t^{(k)} - P_{t-1}^{(k)}\ $; increase t to 50 or add early stopping.
BERT embeddings OOM on CPU (Exp 5)	Medium	Pre-compute on GPU machine; cache .rds; use sentence-transformers quantized model as fallback.
ARI-based weight learning overfits to small datasets	Medium	Use nested CV (inner loop for weight tuning); report both inner and outer loop ARI.
Multiplex Leiden does not improve over A on any dataset	Low–Medium	Constitutes a negative result; include as evidence that C-Multiplex is dataset-specific.

12 Run Schedule

Phase	Prerequisite	Task	Est.
P0	—	Implement <code>er_normalize()</code> , <code>er_classify()</code> , <code>er_snf()</code>	1–2 wk
P1	P0	Exp 2: field ablation on CORA (fast; validates field diagnostics)	1 day
P2	P0	Exp 4: missingness robustness on DBLP-ACM	1 day
P3	P0	Exp 1: main accuracy comparison, all 4 datasets, parallel	1–2 days
P4	P0	Exp 3: scalability sweep on NCVR (n up to 100K)	1 day
P5	P3	Implement <code>er_bow()</code> , <code>er_embed()</code> ; run Exp 5	3–5 days
P6	P1–P5	Meta-analysis (RQ6); write paper Sections 6–7	1 week

13 Summary

This handbook operationalizes the seven selection rules of the ERBOT Method Selection Guide (Slide 15) into five reproducible experiments. The central thread running through all experiments is:

A fusion strategy should be chosen based on the structural properties of the field set — noise, orthogonality, missingness, scale, and dataset size — not on a blind search over all methods. The ERBOT pipeline enforces an explicit, statistically grounded classification step before clustering, ensuring that the match/non-match boundary is always auditable and ER-grounded.

Experiments 1–5, run on CORA, Restaurant, DBLP-ACM, and NCVR, will yield a 7×3 evidence table that either confirms or refutes each selection rule. The meta-analysis (RQ6) will test whether field-level diagnostics (cardinality, missingness, per-field ARI) are sufficient to predict the winning strategy — the strongest form of the paper’s claim.

References

- A. Bagga and B. Baldwin. Algorithms for scoring coreference chains. In *Proceedings of the Linguistic Coreference Workshop at LREC*, pages 563–566, 1998.
- V. D. Blondel, J.-L. Guillaume, R. Lambiotte, and E. Lefebvre. Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008(10):P10008, 2008. doi: 10.1088/1742-5468/2008/10/P10008.
- P. Christen. *Data Matching: Concepts and Techniques for Record Linkage, Entity Resolution, and Duplicate Detection*. Springer, Berlin, 2012. doi: 10.1007/978-3-642-31164-2.
- I. P. Fellegi and A. B. Sunter. A theory for record linkage. *Journal of the American Statistical Association*, 64(328):1183–1210, 1969. doi: 10.1080/01621459.1969.10501049.
- L. Hubert and P. Arabie. Comparing partitions. *Journal of Classification*, 2(1):193–218, 1985. doi: 10.1007/BF01908075.
- G. R. G. Lanckriet, N. Cristianini, P. Bartlett, L. El Ghaoui, and M. I. Jordan. Learning the kernel matrix with semidefinite programming. *Journal of Machine Learning Research*, 5:27–72, 2004.
- P. J. Mucha, T. Richardson, K. Macon, M. A. Porter, and J.-P. Onnela. Community structure in time-dependent, multiscale, and multiplex networks. *Science*, 328(5980):876–878, 2010.
- G. Papadakis, E. Ioannou, E. Thanos, and T. Palpanas. Blocking and filtering techniques for entity resolution: A survey. *ACM Computing Surveys*, 53(2):1–42, 2020. doi: 10.1145/3377455.
- A. Rosenberg and J. Hirschberg. V-measure: A conditional entropy-based external cluster evaluation measure. In *Proceedings of the 2007 Joint Conference on Empirical Methods in Natural Language Processing and Computational Natural Language Learning*, pages 410–420, 2007.
- D. Selivanov, M. Bickel, and Q. Wang. *text2vec: Modern Text Mining Framework for R*, 2020. URL <https://CRAN.R-project.org/package=text2vec>. R package version 0.6.0.
- V. A. Traag, L. Waltman, and N. J. van Eck. From Louvain to Leiden: Guaranteeing well-connected communities. *Scientific Reports*, 9(1):5233, 2019. doi: 10.1038/s41598-019-41695-z.
- N. X. Vinh, J. Epps, and J. Bailey. Information theoretic measures for clusterings comparison: Variants, properties, normalization and correction for chance. *Journal of Machine Learning Research*, 11:2837–2854, 2010.
- B. Wang, A. M. Mezlini, F. Demir, D. Tarlow, G. Taylor, G. D. Bader, and A. Goldenberg. Similarity network fusion for aggregating data types on a genomic scale. *Nature Methods*, 11(3):333–337, 2014.
- W. E. Winkler. String comparator metrics and enhanced decision rules in the Fellegi-Sunter model of record linkage. *Proceedings of the Section on Survey Research Methods*, pages 354–359, 1990.